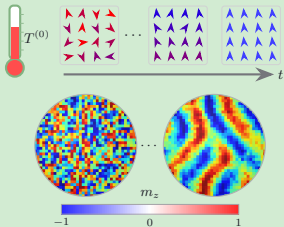


1. Data Generation

Metropolis–Hastings Monte Carlo
Thermodynamically relaxed
2D magnetization fields



Dataset covering diverse
Hamiltonian configurations

Hamiltonian parameters

$$\tilde{J}_1, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4$$
$$\tilde{K}_{an1}, \tilde{K}_{anS}, \tilde{H}_{ex}, \tilde{K}_{DM}$$

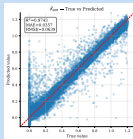
2. Inverse Estimation

Predict Hamiltonian parameters
from 2D textures



Dataset

DL Model

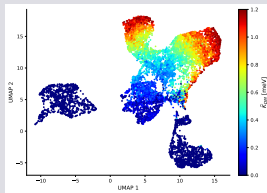


**Predicted
Parameters**

3. Interpretability

Visually and quantitatively connect
latent behavior with physical phenomena

a) Latent space and physical insight



b) Regression Activation Maps (RAMs)

